

Complex Numbers, Circular Functions and Hyperbolic Functions

Part 6 of the OT Science Formula Completion Workbook

OT Science Notes

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1 Complex Numbers, Circular Functions and Hyperbolic Functions

Complex numbers connect algebra, geometry, oscillations, rotations, exponentials and hyperbolic functions. The core bridge is Euler's formula. Once that is understood, trigonometric and hyperbolic identities become different faces of the same exponential structure.

1.1 Basic Complex Algebra

A complex number has a real part and an imaginary part:

$$z = x + iy, \quad i^2 = -1.$$

The complex plane treats x as the horizontal coordinate and y as the vertical coordinate.

Complex Number Basics

$$z = x + iy, \quad \operatorname{Re}(z) = x, \quad \operatorname{Im}(z) = y.$$

$$\bar{z} = x - iy.$$

$$|z| = \sqrt{x^2 + y^2}, \quad \arg(z) = \theta.$$

$$z\bar{z} = |z|^2.$$

$$\frac{1}{z} = \frac{\bar{z}}{|z|^2} = \frac{x - iy}{x^2 + y^2}, \quad z \neq 0.$$

1.2 Polar and Exponential Forms

The polar form is often the best way to multiply, divide, raise to powers and extract roots.

Euler, Polar Form and De Moivre

$$e^{i\theta} = \cos \theta + i \sin \theta.$$

$$z = r(\cos \theta + i \sin \theta) = re^{i\theta}.$$

$$r = |z|, \quad \theta = \arg(z).$$

$$z_1 z_2 = r_1 r_2 e^{i(\theta_1 + \theta_2)}.$$

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} e^{i(\theta_1 - \theta_2)}, \quad z_2 \neq 0.$$

$$(re^{i\theta})^n = r^n e^{in\theta}.$$

$$(\cos \theta + i \sin \theta)^n = \cos(n\theta) + i \sin(n\theta).$$

1.3 Roots of Complex Numbers

If $z^n = Re^{i\Theta}$, then the n roots are evenly spaced around a circle in the complex plane.

Complex Roots

$$z_k = R^{1/n} e^{i(\Theta+2\pi k)/n}, \quad k = 0, 1, 2, \dots, n-1.$$

Equivalently,

$$z_k = R^{1/n} \left[\cos \left(\frac{\Theta + 2\pi k}{n} \right) + i \sin \left(\frac{\Theta + 2\pi k}{n} \right) \right].$$

1.4 Complex Exponentials and Circular Functions

The ordinary circular functions can be written using complex exponentials:

Circular Functions from Exponentials

$$\cos x = \frac{e^{ix} + e^{-ix}}{2}.$$

$$\sin x = \frac{e^{ix} - e^{-ix}}{2i}.$$

$$\tan x = \frac{\sin x}{\cos x}.$$

$$\cos^2 x + \sin^2 x = 1.$$

$$e^{ix} e^{-ix} = 1.$$

1.5 Hyperbolic Functions

Hyperbolic functions are built from real exponentials. They resemble circular functions, but their signs differ in crucial places.

Hyperbolic Definitions and Identities

$$\sinh x = \frac{e^x - e^{-x}}{2}.$$

$$\cosh x = \frac{e^x + e^{-x}}{2}.$$

$$\tanh x = \frac{\sinh x}{\cosh x}.$$

$$\cosh^2 x - \sinh^2 x = 1.$$

$$\sinh(2x) = 2 \sinh x \cosh x.$$

$$\cosh(2x) = \cosh^2 x + \sinh^2 x = 2 \cosh^2 x - 1 = 1 + 2 \sinh^2 x.$$

$$\frac{d}{dx} \sinh x = \cosh x, \quad \frac{d}{dx} \cosh x = \sinh x, \quad \frac{d}{dx} \tanh x = \operatorname{sech}^2 x.$$

1.6 Circular—Hyperbolic Connections

Replacing a real argument by an imaginary one converts circular functions into hyperbolic functions and vice versa.

Complex Arguments and Hyperbolic Connections

$$\cos(ix) = \cosh x.$$

$$\sin(ix) = i \sinh x.$$

$$\tan(ix) = i \tanh x.$$

$$\cosh(ix) = \cos x.$$

$$\sinh(ix) = i \sin x.$$

$$\tanh(ix) = i \tan x.$$

1.7 Inverse Hyperbolic Functions

Inverse hyperbolic functions can be written using logarithms. These are important in integration and differential equations.

Inverse Hyperbolic Forms

$$\sinh^{-1} x = \ln(x + \sqrt{x^2 + 1}).$$

$$\cosh^{-1} x = \ln(x + \sqrt{x^2 - 1}), \quad x \geq 1.$$

$$\tanh^{-1} x = \frac{1}{2} \ln\left(\frac{1+x}{1-x}\right), \quad |x| < 1.$$

Quick Drills and Practice: Complex and Hyperbolic Functions

Level A: Formula recall

1. Complete: $e^{i\theta} =$ _____.
2. Complete: $z\bar{z} =$ _____.
3. Complete: $1/(x + iy) =$ _____.
4. Complete: $\cos x =$ _____.
5. Complete: $\sin x =$ _____.
6. Complete: $\sinh x =$ _____.
7. Complete: $\cosh x =$ _____.
8. Complete: $\cosh^2 x - \sinh^2 x =$ _____.
9. Complete: $\cos(ix) =$ _____.
10. Complete: $\sin(ix) =$ _____.

Level B: Complex algebra

1. Compute $(3 + 2i) + (1 - 5i)$.
2. Compute $(2 + i)(3 - 4i)$.
3. Compute $(1 + i)^4$ using ordinary algebra.
4. Write $1 + i$ in polar and exponential form.
5. Write $-1 + \sqrt{3}i$ in polar and exponential form.
6. Compute $(1 + i)^8$ using polar form.
7. Compute $\frac{2+3i}{1-i}$.

Level C: Powers and roots

1. Find all cube roots of 8.
2. Find all fourth roots of 1.
3. Find all fourth roots of -16 .
4. Solve $z^3 = -8i$.
5. Sketch the roots of $z^5 = 1$ on the complex plane.

Level D: Hyperbolic connections

1. Prove $\cos(ix) = \cosh x$ using exponential definitions.
2. Prove $\sin(ix) = i \sinh x$.
3. Prove $\cosh(ix) = \cos x$.
4. Prove $\sinh(ix) = i \sin x$.
5. Prove $\cosh^2 x - \sinh^2 x = 1$.
6. Derive $\sinh(2x) = 2 \sinh x \cosh x$.
7. Derive $\cosh(2x) = \cosh^2 x + \sinh^2 x$.

Level E: Challenge

1. Express $\cos(x + iy)$ in terms of $\cos x, \sin x, \cosh y, \sinh y$.
2. Express $\sin(x + iy)$ in terms of $\cos x, \sin x, \cosh y, \sinh y$.
3. Solve $e^z = -1$ for complex z .
4. Solve $\cos z = 0$ for complex z .
5. Show that multiplication by $e^{i\theta}$ rotates a complex number by angle θ .